

Safe undo

You will edit the wrong file, stage too early, or want to back out of a local mistake

Restore before reset

- To drop unstaged edits in a tracked file
- To unstage without losing your edits
- Soft reset moves HEAD back while keeping your staged work; mixed reset also unstages
- Hard reset throws away local changes, save that for when you truly mean it

Browser lab

Digital Design and Verification Platform

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Tools

INTERACTIVE TOOL

Safe Git undo

Practice **unstage**, **restore**, and **soft reset** before touching rewrite commands. See working tree ↔ index ↔ HEAD update live.

Starter example: rtl/top.v is staged broken *and* has extra worktree edits; notes.txt has only unstaged edits. Unstage / restore safely — avoid hard reset unless you mean it.

Load starter example

Challenges 0 / 22 cleared

Quiz: unstage: To unstage without losing edits, use? Answer: restore --staged

Answer

Show hintCheckNextIdle

Quiz: unstageQuiz: layersUnstage topQuiz: restoreRestore notesQuiz: softSoft resetQuiz: mixedMixed resetQuiz: hardHard gatedHard resetQuiz: forceForce blockedPrefer softUnstage keeps workClean top safelyQuiz: old habitQuiz: path unstageSoft then fixStarter depthHabit check

Three layers

HEAD e4f5a6b “wire alu into top” · parent a1b2c3d · not pushed

rtl/top.v staged+dirtynotes.txt dirty

green border = matches neighbor amber = differs from HEAD

HEAD (commit)
e4f5a6b · wire alu into top

Index (staged)
differs from HEAD

Working tree
uncommitted edits

Undo actions

Safe zone

Keeps or moves changes without deleting history.

git restore --staged <file>

git restore <file>

Danger zone

Can discard work or rewrite published history.

☐ I understand this can discard work

git reset --hard HEAD~1

Safe Git undo lab starter

learn_git — git undo safe

Real Git practice

```
wsl - module06-git-undo-safe/examples/remote_undo

# Track A - real Unix
# real shell session (Track A)
# practice repo: /tmp/git-undo-KtVZGh

$ git init

$ echo "v1" > notes.md

$ git add notes.md

$ git commit -m "Add notes"

$ echo "draft line" >> notes.md

$ git status
On branch master
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   notes.md

no changes added to commit (use "git add" and/or "git commit -a")

$ git restore notes.md

$ echo "draft line" >> notes.md

$ git add notes.md

$ git restore --staged notes.md

$ git status
On branch master
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   notes.md

no changes added to commit (use "git add" and/or "git commit -a")
```

Real shell — restore and unstage

Real Git practice — try these

```
# git init — create a new repository here  
git init
```

```
# echo ... > notes.md — create and commit a tracked file  
echo "v1" > notes.md
```

```
git add notes.md
```

```
git commit -m "Add notes"
```

```
# echo ... >> notes.md — edit without staging  
echo "draft line" >> notes.md
```

```
# git status — see modified, not staged  
git status
```

Pitfalls to watch

- Do not confuse unstage with discard, restore staged keeps your edits
- Avoid reset hard until you have tried restore and soft or mixed reset
- And remember

Your turn

- Complete the checklist for at least one track, preferably both
- In the browser, load the starter and practice unstage versus restore on a few challenges
- On real Git, repeat restore and restore staged in your practice tree
- When you are ready